

# Everything about DSA — 56 Patterns × LeetCode Questions

10 Categories | 56 Patterns | 5-10 Questions per Pattern with LeetCode Links

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## 1. Arrays (5 Patterns)

### 1.1 Two Pointers

1. Two Sum II - Input Array Is Sorted — <https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/>
2. 3Sum — <https://leetcode.com/problems/3sum/>
3. Container With Most Water — <https://leetcode.com/problems/container-with-most-water/>
4. Sort Colors (Dutch National Flag) — <https://leetcode.com/problems/sort-colors/>
5. Remove Duplicates from Sorted Array — <https://leetcode.com/problems/remove-duplicates-from-sorted-array/>
6. 4Sum — <https://leetcode.com/problems/4sum/>
7. Trapping Rain Water — <https://leetcode.com/problems/trapping-rain-water/>

### 1.2 Merge Intervals

1. Merge Intervals — <https://leetcode.com/problems/merge-intervals/>
2. Insert Interval — <https://leetcode.com/problems/insert-interval/>
3. Non-overlapping Intervals — <https://leetcode.com/problems/non-overlapping-intervals/>
4. Meeting Rooms II — <https://leetcode.com/problems/meeting-rooms-ii/>
5. Interval List Intersections — <https://leetcode.com/problems/interval-list-intersections/>
6. Minimum Number of Arrows to Burst Balloons — <https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/>
7. Employee Free Time — <https://leetcode.com/problems/employee-free-time/>

### 1.3 Sorting

1. Merge Sorted Array — <https://leetcode.com/problems/merge-sorted-array/>
2. Sort an Array — <https://leetcode.com/problems/sort-an-array/>

3. Largest Number — <https://leetcode.com/problems/largest-number/>
4. Kth Largest Element in an Array — <https://leetcode.com/problems/kth-largest-element-in-an-array/>
5. Relative Sort Array — <https://leetcode.com/problems/relative-sort-array/>
6. Sort Array By Parity — <https://leetcode.com/problems/sort-array-by-parity/>
7. H-Index — <https://leetcode.com/problems/h-index/>

## 1.4 Sliding Window

1. Maximum Subarray — <https://leetcode.com/problems/maximum-subarray/>
2. Maximum Average Subarray I — <https://leetcode.com/problems/maximum-average-subarray-i/>
3. Longest Substring Without Repeating Characters — <https://leetcode.com/problems/longest-substring-without-repeating-characters/>
4. Minimum Size Subarray Sum — <https://leetcode.com/problems/minimum-size-subarray-sum/>
5. Sliding Window Maximum — <https://leetcode.com/problems/sliding-window-maximum/>
6. Fruit Into Baskets — <https://leetcode.com/problems/fruit-into-baskets/>
7. Max Consecutive Ones III — <https://leetcode.com/problems/max-consecutive-ones-iii/>

## 1.5 Prefix Sums

1. Range Sum Query - Immutable — <https://leetcode.com/problems/range-sum-query-immutable/>
  2. Subarray Sum Equals K — <https://leetcode.com/problems/subarray-sum-equals-k/>
  3. Product of Array Except Self — <https://leetcode.com/problems/product-of-array-except-self/>
  4. Continuous Subarray Sum — <https://leetcode.com/problems/continuous-subarray-sum/>
  5. Find Pivot Index — <https://leetcode.com/problems/find-pivot-index/>
  6. Maximum Size Subarray Sum Equals k — <https://leetcode.com/problems/maximum-size-subarray-sum-equals-k/>
  7. Range Sum Query 2D - Immutable — <https://leetcode.com/problems/range-sum-query-2d-immutable/>
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## 2. Strings (5 Patterns)

### 2.1 Two Pointers

1. Valid Palindrome — <https://leetcode.com/problems/valid-palindrome/>
2. Reverse String — <https://leetcode.com/problems/reverse-string/>
3. Reverse Words in a String — <https://leetcode.com/problems/reverse-words-in-a-string/>
4. Palindromic Substrings — <https://leetcode.com/problems/palindromic-substrings/>
5. Is Subsequence — <https://leetcode.com/problems/is-subsequence/>
6. Backspace String Compare — <https://leetcode.com/problems/backspace-string-compare/>

### 2.2 Sliding Window

1. Longest Substring Without Repeating Characters — <https://leetcode.com/problems/longest-substring-without-repeating-characters/>
2. Minimum Window Substring — <https://leetcode.com/problems/minimum-window-substring/>
3. Longest Repeating Character Replacement — <https://leetcode.com/problems/longest-repeating-character-replacement/>
4. Find All Anagrams in a String — <https://leetcode.com/problems/find-all-anagrams-in-a-string/>
5. Permutation in String — <https://leetcode.com/problems/permutation-in-string/>
6. Longest Substring with At Most K Distinct Characters — <https://leetcode.com/problems/longest-substring-with-at-most-k-distinct-characters/>

### 2.3 Hashmaps

1. Two Sum — <https://leetcode.com/problems/two-sum/>
2. Group Anagrams — <https://leetcode.com/problems/group-anagrams/>
3. Valid Anagram — <https://leetcode.com/problems/valid-anagram/>
4. Ransom Note — <https://leetcode.com/problems/ransom-note/>
5. Longest Consecutive Sequence — <https://leetcode.com/problems/longest-consecutive-sequence/>
6. Isomorphic Strings — <https://leetcode.com/problems/isomorphic-strings/>
7. First Unique Character in a String — <https://leetcode.com/problems/first-unique-character-in-a-string/>

### 2.4 String Manipulation

1. String to Integer (atoi) — <https://leetcode.com/problems/string-to-integer-atoi/>

2. Implement strStr() — <https://leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string/>
3. Zigzag Conversion — <https://leetcode.com/problems/zigzag-conversion/>
4. Multiply Strings — <https://leetcode.com/problems/multiply-strings/>
5. Compare Version Numbers — <https://leetcode.com/problems/compare-version-numbers/>
6. Encode and Decode Strings — <https://leetcode.com/problems/encode-and-decode-strings/>

## 2.5 Regular Expressions

1. Regular Expression Matching — <https://leetcode.com/problems/regular-expression-matching/>
  2. Wildcard Matching — <https://leetcode.com/problems/wildcard-matching/>
  3. Valid Number — <https://leetcode.com/problems/valid-number/>
  4. Basic Calculator — <https://leetcode.com/problems/basic-calculator/>
  5. Valid Parenthesis String — <https://leetcode.com/problems/valid-parenthesis-string/>
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## 3. Binary Search (5 Patterns)

### 3.1 Basic Binary Search

1. Binary Search — <https://leetcode.com/problems/binary-search/>
2. Search Insert Position — <https://leetcode.com/problems/search-insert-position/>
3. Sqrt(x) — <https://leetcode.com/problems/sqrtx/>
4. Guess Number Higher or Lower — <https://leetcode.com/problems/guess-number-higher-or-lower/>
5. First Bad Version — <https://leetcode.com/problems/first-bad-version/>
6. Valid Perfect Square — <https://leetcode.com/problems/valid-perfect-square/>

### 3.2 Range Search

1. Find First and Last Position of Element in Sorted Array — <https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/>
2. Search in Rotated Sorted Array — <https://leetcode.com/problems/search-in-rotated-sorted-array/>
3. Search in Rotated Sorted Array II — <https://leetcode.com/problems/search-in-rotated-sorted-array-ii/>
4. Find Minimum in Rotated Sorted Array — <https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/>

5. Find K Closest Elements — <https://leetcode.com/problems/find-k-closest-elements/>
6. Time Based Key-Value Store — <https://leetcode.com/problems/time-based-key-value-store/>

### 3.3 Allocation / Capacity Problems (Binary Search on Answer)

1. Koko Eating Bananas — <https://leetcode.com/problems/koko-eating-bananas/>
2. Capacity To Ship Packages Within D Days — <https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/>
3. Split Array Largest Sum — <https://leetcode.com/problems/split-array-largest-sum/>
4. Allocate Minimum Number of Pages (GFG) — <https://www.geeksforgeeks.org/allocate-minimum-number-pages/>
5. Minimum Number of Days to Make m Bouquets — <https://leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets/>
6. Magnetic Force Between Two Balls — <https://leetcode.com/problems/magnetic-force-between-two-balls/>

### 3.4 Counting Occurrences

1. Find First and Last Position of Element in Sorted Array — <https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/>
2. Count Negative Numbers in a Sorted Matrix — <https://leetcode.com/problems/count-negative-numbers-in-a-sorted-matrix/>
3. Kth Missing Positive Number — <https://leetcode.com/problems/kth-missing-positive-number/>
4. Single Element in a Sorted Array — <https://leetcode.com/problems/single-element-in-a-sorted-array/>
5. Count of Smaller Numbers After Self — <https://leetcode.com/problems/count-of-smaller-numbers-after-self/>

### 3.5 Bitonic Array Search

1. Peak Index in a Mountain Array — <https://leetcode.com/problems/peak-index-in-a-mountain-array/>
  2. Find in Mountain Array — <https://leetcode.com/problems/find-in-mountain-array/>
  3. Find Peak Element — <https://leetcode.com/problems/find-peak-element/>
  4. Valid Mountain Array — <https://leetcode.com/problems/valid-mountain-array/>
  5. Minimum Number of Removals to Make Mountain Array — <https://leetcode.com/problems/minimum-number-of-removals-to-make-mountain-array/>
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## 4. Recursion (4 Patterns)

### 4.1 Basic Recursive Functions

1. Fibonacci Number — <https://leetcode.com/problems/fibonacci-number/>
2. Reverse Linked List (recursive) — <https://leetcode.com/problems/reverse-linked-list/>
3. Power of x to n ( $\text{Pow}(x, n)$ ) — <https://leetcode.com/problems/powx-n/>
4. Climbing Stairs — <https://leetcode.com/problems/climbing-stairs/>
5. Sum of Left Leaves — <https://leetcode.com/problems/sum-of-left-leaves/>
6. Factorial Trailing Zeroes — <https://leetcode.com/problems/factorial-trailing-zeroes/>

### 4.2 Divide & Conquer

1. Maximum Subarray — <https://leetcode.com/problems/maximum-subarray/>
2. Merge Sort based: Sort an Array — <https://leetcode.com/problems/sort-an-array/>
3. Kth Largest Element in an Array — <https://leetcode.com/problems/kth-largest-element-in-an-array/>
4. Different Ways to Add Parentheses — <https://leetcode.com/problems/different-ways-to-add-parentheses/>
5. The Skyline Problem — <https://leetcode.com/problems/the-skyline-problem/>
6. Count of Smaller Numbers After Self — <https://leetcode.com/problems/count-of-smaller-numbers-after-self/>

### 4.3 Backtracking

1. Permutations — <https://leetcode.com/problems/permutations/>
2. Subsets — <https://leetcode.com/problems/subsets/>
3. Combination Sum — <https://leetcode.com/problems/combination-sum/>
4. N-Queens — <https://leetcode.com/problems/n-queens/>
5. Word Search — <https://leetcode.com/problems/word-search/>
6. Palindrome Partitioning — <https://leetcode.com/problems/palindrome-partitioning/>
7. Sudoku Solver — <https://leetcode.com/problems/sudoku-solver/>
8. Letter Combinations of a Phone Number — <https://leetcode.com/problems/letter-combinations-of-a-phone-number/>

### 4.4 Recursive Search

1. Binary Tree Inorder Traversal — <https://leetcode.com/problems/binary-tree-inorder-traversal/>
2. Path Sum — <https://leetcode.com/problems/path-sum/>
3. Validate Binary Search Tree — <https://leetcode.com/problems/validate-binary-search-tree/>

4. Number of Islands (DFS) — <https://leetcode.com/problems/number-of-islands/>
  5. Word Break — <https://leetcode.com/problems/word-break/>
  6. Generate Parentheses — <https://leetcode.com/problems/generate-parentheses/>
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## 5. Linked List (4 Patterns)

### 5.1 Fast and Slow Pointers

1. Linked List Cycle — <https://leetcode.com/problems/linked-list-cycle/>
2. Linked List Cycle II — <https://leetcode.com/problems/linked-list-cycle-ii/>
3. Middle of the Linked List — <https://leetcode.com/problems/middle-of-the-linked-list/>
4. Palindrome Linked List — <https://leetcode.com/problems/palindrome-linked-list/>
5. Happy Number — <https://leetcode.com/problems/happy-number/>
6. Reorder List — <https://leetcode.com/problems/reorder-list/>
7. Remove Nth Node From End of List — <https://leetcode.com/problems/remove-nth-node-from-end-of-list/>

### 5.2 Dummy Node Technique

1. Merge Two Sorted Lists — <https://leetcode.com/problems/merge-two-sorted-lists/>
2. Remove Duplicates from Sorted List II — <https://leetcode.com/problems/remove-duplicates-from-sorted-list-ii/>
3. Partition List — <https://leetcode.com/problems/partition-list/>
4. Add Two Numbers — <https://leetcode.com/problems/add-two-numbers/>
5. Swap Nodes in Pairs — <https://leetcode.com/problems/swap-nodes-in-pairs/>
6. Merge k Sorted Lists — <https://leetcode.com/problems/merge-k-sorted-lists/>

### 5.3 Recursion Technique

1. Reverse Linked List — <https://leetcode.com/problems/reverse-linked-list/>
2. Swap Nodes in Pairs — <https://leetcode.com/problems/swap-nodes-in-pairs/>
3. Merge Two Sorted Lists — <https://leetcode.com/problems/merge-two-sorted-lists/>
4. Flatten a Multilevel Doubly Linked List — <https://leetcode.com/problems/flatten-a-multilevel-doubly-linked-list/>
5. Copy List with Random Pointer — <https://leetcode.com/problems/copy-list-with-random-pointer/>

### 5.4 In-Place Reversal Technique

1. Reverse Linked List — <https://leetcode.com/problems/reverse-linked-list/>
2. Reverse Linked List II — <https://leetcode.com/problems/reverse-linked-list-ii/>

3. Reverse Nodes in k-Group — <https://leetcode.com/problems/reverse-nodes-in-k-group/>
  4. Rotate List — <https://leetcode.com/problems/rotate-list/>
  5. Swapping Nodes in a Linked List — <https://leetcode.com/problems/swapping-nodes-in-a-linked-list/>
  6. Odd Even Linked List — <https://leetcode.com/problems/odd-even-linked-list/>
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## 6. Stacks & Queues (5 Patterns)

### 6.1 Design Problems

1. Min Stack — <https://leetcode.com/problems/min-stack/>
2. Implement Queue using Stacks — <https://leetcode.com/problems/implement-queue-using-stacks/>
3. Implement Stack using Queues — <https://leetcode.com/problems/implement-stack-using-queues/>
4. Design Circular Queue — <https://leetcode.com/problems/design-circular-queue/>
5. LRU Cache — <https://leetcode.com/problems/lru-cache/>
6. Design a Stack With Increment Operation — <https://leetcode.com/problems/design-a-stack-with-increment-operation/>

### 6.2 Monotonic Stack

1. Daily Temperatures — <https://leetcode.com/problems/daily-temperatures/>
2. Next Greater Element I — <https://leetcode.com/problems/next-greater-element-i/>
3. Largest Rectangle in Histogram — <https://leetcode.com/problems/largest-rectangle-in-histogram/>
4. Trapping Rain Water — <https://leetcode.com/problems/trapping-rain-water/>
5. Remove K Digits — <https://leetcode.com/problems/remove-k-digits/>
6. Online Stock Span — <https://leetcode.com/problems/online-stock-span/>
7. Sum of Subarray Minimums — <https://leetcode.com/problems/sum-of-subarray-minimums/>

### 6.3 Expression Evaluation

1. Valid Parentheses — <https://leetcode.com/problems/valid-parentheses/>
2. Evaluate Reverse Polish Notation — <https://leetcode.com/problems/evaluate-reverse-polish-notation/>
3. Basic Calculator — <https://leetcode.com/problems/basic-calculator/>
4. Basic Calculator II — <https://leetcode.com/problems/basic-calculator-ii/>
5. Decode String — <https://leetcode.com/problems/decode-string/>



6. Remove All Adjacent Duplicates in String — <https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string/>

## 6.4 Two Stacks

1. Min Stack — <https://leetcode.com/problems/min-stack/>
2. Implement Queue using Stacks — <https://leetcode.com/problems/implement-queue-using-stacks/>
3. Max Stack — <https://leetcode.com/problems/max-stack/>
4. Design a Stack With Increment Operation — <https://leetcode.com/problems/design-a-stack-with-increment-operation/>
5. Simplify Path — <https://leetcode.com/problems/simplify-path/>

## 6.5 Sliding Window & Monotonic Queue

1. Sliding Window Maximum — <https://leetcode.com/problems/sliding-window-maximum/>
  2. Shortest Subarray with Sum at Least K — <https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/>
  3. Constrained Subsequence Sum — <https://leetcode.com/problems/constrained-subsequence-sum/>
  4. Jump Game VI — <https://leetcode.com/problems/jump-game-vi/>
  5. Sum of Subarray Ranges — <https://leetcode.com/problems/sum-of-subarray-ranges/>
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## 7. Binary Trees & BST (6 Patterns)

### 7.1 Traversal

1. Binary Tree Inorder Traversal — <https://leetcode.com/problems/binary-tree-inorder-traversal/>
2. Binary Tree Preorder Traversal — <https://leetcode.com/problems/binary-tree-preorder-traversal/>
3. Binary Tree Postorder Traversal — <https://leetcode.com/problems/binary-tree-postorder-traversal/>
4. Binary Tree Level Order Traversal — <https://leetcode.com/problems/binary-tree-level-order-traversal/>
5. Binary Tree Zigzag Level Order Traversal — <https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/>
6. Vertical Order Traversal of a Binary Tree — <https://leetcode.com/problems/vertical-order-traversal-of-a-binary-tree/>

### 7.2 Construction

1. Construct Binary Tree from Preorder and Inorder Traversal — <https://leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal/>
2. Construct Binary Tree from Inorder and Postorder Traversal — <https://leetcode.com/problems/construct-binary-tree-from-inorder-and-postorder-traversal/>
3. Convert Sorted Array to Binary Search Tree — <https://leetcode.com/problems/convert-sorted-array-to-binary-search-tree/>
4. Serialize and Deserialize Binary Tree — <https://leetcode.com/problems/serialize-and-deserialize-binary-tree/>
5. Construct Binary Search Tree from Preorder Traversal — <https://leetcode.com/problems/construct-binary-search-tree-from-preorder-traversal/>
6. Maximum Binary Tree — <https://leetcode.com/problems/maximum-binary-tree/>

### 7.3 Mirror & Symmetry

1. Symmetric Tree — <https://leetcode.com/problems/symmetric-tree/>
2. Invert Binary Tree — <https://leetcode.com/problems/invert-binary-tree/>
3. Same Tree — <https://leetcode.com/problems/same-tree/>
4. Subtree of Another Tree — <https://leetcode.com/problems/subtree-of-another-tree/>
5. Flip Equivalent Binary Trees — <https://leetcode.com/problems/flip-equivalent-binary-trees/>

### 7.4 Path Sum & Root to Leaf

1. Path Sum — <https://leetcode.com/problems/path-sum/>
2. Path Sum II — <https://leetcode.com/problems/path-sum-ii/>
3. Path Sum III — <https://leetcode.com/problems/path-sum-iii/>
4. Binary Tree Maximum Path Sum — <https://leetcode.com/problems/binary-tree-maximum-path-sum/>
5. Sum Root to Leaf Numbers — <https://leetcode.com/problems/sum-root-to-leaf-numbers/>
6. Binary Tree Paths — <https://leetcode.com/problems/binary-tree-paths/>

### 7.5 Traversal & Search

1. Lowest Common Ancestor of a Binary Tree — <https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree/>
2. Lowest Common Ancestor of a Binary Search Tree — <https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/>
3. Kth Smallest Element in a BST — <https://leetcode.com/problems/kth-smallest-element-in-a-bst/>

4. Search in a Binary Search Tree — <https://leetcode.com/problems/search-in-a-binary-search-tree/>
5. Binary Search Tree Iterator — <https://leetcode.com/problems/binary-search-tree-iterator/>
6. Closest Binary Search Tree Value — <https://leetcode.com/problems/closest-binary-search-tree-value/>

## 7.6 Validation & Properties

1. Validate Binary Search Tree — <https://leetcode.com/problems/validate-binary-search-tree/>
  2. Balanced Binary Tree — <https://leetcode.com/problems/balanced-binary-tree/>
  3. Maximum Depth of Binary Tree — <https://leetcode.com/problems/maximum-depth-of-binary-tree/>
  4. Minimum Depth of Binary Tree — <https://leetcode.com/problems/minimum-depth-of-binary-tree/>
  5. Count Complete Tree Nodes — <https://leetcode.com/problems/count-complete-tree-nodes/>
  6. Diameter of Binary Tree — <https://leetcode.com/problems/diameter-of-binary-tree/>
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## 8. Priority Queues / Heaps (7 Patterns)

### 8.1 Finding Kth Largest/Smallest

1. Kth Largest Element in an Array — <https://leetcode.com/problems/kth-largest-element-in-an-array/>
2. Kth Largest Element in a Stream — <https://leetcode.com/problems/kth-largest-element-in-a-stream/>
3. K Closest Points to Origin — <https://leetcode.com/problems/k-closest-points-to-origin/>
4. Find K Pairs with Smallest Sums — <https://leetcode.com/problems/find-k-pairs-with-smallest-sums/>
5. Kth Smallest Element in a Sorted Matrix — <https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/>
6. Kth Largest Element in a BST — <https://leetcode.com/problems/kth-largest-element-in-a-bst/>

### 8.2 Top K Frequent Elements

1. Top K Frequent Elements — <https://leetcode.com/problems/top-k-frequent-elements/>
2. Top K Frequent Words — <https://leetcode.com/problems/top-k-frequent-words/>

3. Sort Characters By Frequency — <https://leetcode.com/problems/sort-characters-by-frequency/>
4. Task Scheduler — <https://leetcode.com/problems/task-scheduler/>
5. Rearrange String k Distance Apart — <https://leetcode.com/problems/rearrange-string-k-distance-apart/>

### 8.3 Merge K Lists

1. Merge k Sorted Lists — <https://leetcode.com/problems/merge-k-sorted-lists/>
2. Smallest Range Covering Elements from K Lists — <https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/>
3. Find K Pairs with Smallest Sums — <https://leetcode.com/problems/find-k-pairs-with-smallest-sums/>
4. Merge Sorted Array — <https://leetcode.com/problems/merge-sorted-array/>
5. Ugly Number II — <https://leetcode.com/problems/ugly-number-ii/>

### 8.4 Sliding Window Maximum/Minimum

1. Sliding Window Maximum — <https://leetcode.com/problems/sliding-window-maximum/>
2. Sliding Window Median — <https://leetcode.com/problems/sliding-window-median/>
3. Find Median from Data Stream — <https://leetcode.com/problems/find-median-from-data-stream/>
4. Shortest Subarray with Sum at Least K — <https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/>
5. Constrained Subsequence Sum — <https://leetcode.com/problems/constrained-subsequence-sum/>

### 8.5 Design Problems

1. Design Twitter — <https://leetcode.com/problems/design-twitter/>
2. Find Median from Data Stream — <https://leetcode.com/problems/find-median-from-data-stream/>
3. Maximum Frequency Stack — <https://leetcode.com/problems/maximum-frequency-stack/>
4. Design a Stack With Increment Operation — <https://leetcode.com/problems/design-a-stack-with-increment-operation/>
5. Seat Reservation Manager — <https://leetcode.com/problems/seat-reservation-manager/>

### 8.6 Construction and Manipulation

1. Last Stone Weight — <https://leetcode.com/problems/last-stone-weight/>
2. Reorganize String — <https://leetcode.com/problems/reorganize-string/>

3. Minimum Cost to Connect Sticks — <https://leetcode.com/problems/minimum-cost-to-connect-sticks/>
4. IPO — <https://leetcode.com/problems/ipo/>
5. Furthest Building You Can Reach — <https://leetcode.com/problems/furthest-building-you-can-reach/>

## 8.7 With Graphs

1. Network Delay Time (Dijkstra) — <https://leetcode.com/problems/network-delay-time/>
  2. Path with Maximum Probability — <https://leetcode.com/problems/path-with-maximum-probability/>
  3. Cheapest Flights Within K Stops — <https://leetcode.com/problems/cheapest-flights-within-k-stops/>
  4. Swim in Rising Water — <https://leetcode.com/problems/swim-in-rising-water/>
  5. The Maze II — <https://leetcode.com/problems/the-maze-ii/>
  6. Minimum Cost to Make at Least One Valid Path in a Grid — <https://leetcode.com/problems/minimum-cost-to-make-at-least-one-valid-path-in-a-grid/>
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## 9. Dynamic Programming (8 Patterns)

### 9.1 Basic Dynamic Programming

1. Climbing Stairs — <https://leetcode.com/problems/climbing-stairs/>
2. House Robber — <https://leetcode.com/problems/house-robber/>
3. Min Cost Climbing Stairs — <https://leetcode.com/problems/min-cost-climbing-stairs/>
4. Maximum Subarray — <https://leetcode.com/problems/maximum-subarray/>
5. Fibonacci Number — <https://leetcode.com/problems/fibonacci-number/>
6. Delete and Earn — <https://leetcode.com/problems/delete-and-earn/>

### 9.2 Optimal Substructure

1. Longest Increasing Subsequence — <https://leetcode.com/problems/longest-increasing-subsequence/>
2. Longest Common Subsequence — <https://leetcode.com/problems/longest-common-subsequence/>
3. Edit Distance — <https://leetcode.com/problems/edit-distance/>
4. Maximum Product Subarray — <https://leetcode.com/problems/maximum-product-subarray/>
5. Word Break — <https://leetcode.com/problems/word-break/>
6. Decode Ways — <https://leetcode.com/problems/decode-ways/>

### 9.3 Interval / Range DP

1. Burst Balloons — <https://leetcode.com/problems/burst-balloons/>
2. Palindrome Partitioning II — <https://leetcode.com/problems/palindrome-partitioning-ii/>
3. Minimum Cost Tree From Leaf Values — <https://leetcode.com/problems/minimum-cost-tree-from-leaf-values/>
4. Strange Printer — <https://leetcode.com/problems/strange-printer/>
5. Remove Boxes — <https://leetcode.com/problems/remove-boxes/>
6. Predict the Winner — <https://leetcode.com/problems/predict-the-winner/>

### 9.4 Knapsack Problems

1. Partition Equal Subset Sum — <https://leetcode.com/problems/partition-equal-subset-sum/>
2. Coin Change — <https://leetcode.com/problems/coin-change/>
3. Coin Change II — <https://leetcode.com/problems/coin-change-ii/>
4. Target Sum — <https://leetcode.com/problems/target-sum/>
5. Ones and Zeroes — <https://leetcode.com/problems/ones-and-zeroes/>
6. Last Stone Weight II — <https://leetcode.com/problems/last-stone-weight-ii/>

### 9.5 Prefix Sums (DP variant)

1. Range Sum Query - Immutable — <https://leetcode.com/problems/range-sum-query-immutable/>
2. Maximum Subarray — <https://leetcode.com/problems/maximum-subarray/>
3. Maximum Sum Circular Subarray — <https://leetcode.com/problems/maximum-sum-circular-subarray/>
4. Minimum Falling Path Sum — <https://leetcode.com/problems/minimum-falling-path-sum/>
5. Maximal Square — <https://leetcode.com/problems/maximal-square/>

### 9.6 Counting Problems

1. Unique Paths — <https://leetcode.com/problems/unique-paths/>
2. Unique Paths II — <https://leetcode.com/problems/unique-paths-ii/>
3. Distinct Subsequences — <https://leetcode.com/problems/distinct-subsequences/>
4. Combination Sum IV — <https://leetcode.com/problems/combination-sum-iv/>
5. Number of Ways to Stay in the Same Place After Some Steps — <https://leetcode.com/problems/number-of-ways-to-stay-in-the-same-place-after-some-steps/>

6. Knight Probability in Chessboard — <https://leetcode.com/problems/knight-probability-in-chessboard/>

## 9.7 Interval Partitioning

1. Word Break II — <https://leetcode.com/problems/word-break-ii/>
2. Palindrome Partitioning — <https://leetcode.com/problems/palindrome-partitioning/>
3. Split Array Largest Sum — <https://leetcode.com/problems/split-array-largest-sum/>
4. Minimum Number of Refueling Stops — <https://leetcode.com/problems/minimum-number-of-refueling-stops/>
5. Video Stitching — <https://leetcode.com/problems/video-stitching/>

## 9.8 Probability & Expectations

1. Knight Probability in Chessboard — <https://leetcode.com/problems/knight-probability-in-chessboard/>
  2. New 21 Game — <https://leetcode.com/problems/new-21-game/>
  3. Soup Servings — <https://leetcode.com/problems/soup-servings/>
  4. Airplane Seat Assignment Probability — <https://leetcode.com/problems/airplane-seat-assignment-probability/>
  5. Random Point in Non-overlapping Rectangles — <https://leetcode.com/problems/random-point-in-non-overlapping-rectangles/>
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## 10. Graphs (7 Patterns)

### 10.1 BFS / DFS Traversal

1. Number of Islands — <https://leetcode.com/problems/number-of-islands/>
2. Flood Fill — <https://leetcode.com/problems/flood-fill/>
3. Clone Graph — <https://leetcode.com/problems/clone-graph/>
4. Max Area of Island — <https://leetcode.com/problems/max-area-of-island/>
5. Rotting Oranges — <https://leetcode.com/problems/rotting-oranges/>
6. Walls and Gates — <https://leetcode.com/problems/walls-and-gates/>

### 10.2 Topological Sort

1. Course Schedule — <https://leetcode.com/problems/course-schedule/>
2. Course Schedule II — <https://leetcode.com/problems/course-schedule-ii/>
3. Alien Dictionary — <https://leetcode.com/problems/alien-dictionary/>
4. Minimum Height Trees — <https://leetcode.com/problems/minimum-height-trees/>
5. Sequence Reconstruction — <https://leetcode.com/problems/sequence-reconstruction/>
6. Parallel Courses — <https://leetcode.com/problems/parallel-courses/>

### 10.3 Union Find (Disjoint Set)

1. Number of Provinces — <https://leetcode.com/problems/number-of-provinces/>
2. Redundant Connection — <https://leetcode.com/problems/redundant-connection/>
3. Accounts Merge — <https://leetcode.com/problems/accounts-merge/>
4. Number of Islands II — <https://leetcode.com/problems/number-of-islands-ii/>
5. Graph Valid Tree — <https://leetcode.com/problems/graph-valid-tree/>
6. Satisfiability of Equality Equations — <https://leetcode.com/problems/satisfiability-of-equality-equations/>

### 10.4 Shortest Path (Dijkstra / Bellman-Ford)

1. Network Delay Time — <https://leetcode.com/problems/network-delay-time/>
2. Cheapest Flights Within K Stops — <https://leetcode.com/problems/cheapest-flights-within-k-stops/>
3. Path With Minimum Effort — <https://leetcode.com/problems/path-with-minimum-effort/>
4. Path with Maximum Probability — <https://leetcode.com/problems/path-with-maximum-probability/>
5. Swim in Rising Water — <https://leetcode.com/problems/swim-in-rising-water/>
6. Minimum Cost to Make at Least One Valid Path in a Grid — <https://leetcode.com/problems/minimum-cost-to-make-at-least-one-valid-path-in-a-grid/>

### 10.5 Minimum Spanning Tree (Prim's / Kruskal's)

1. Min Cost to Connect All Points — <https://leetcode.com/problems/min-cost-to-connect-all-points/>
2. Connecting Cities With Minimum Cost — <https://leetcode.com/problems/connecting-cities-with-minimum-cost/>
3. Optimize Water Distribution in a Village — <https://leetcode.com/problems/optimize-water-distribution-in-a-village/>
4. Find Critical and Pseudo-Critical Edges in MST — <https://leetcode.com/problems/find-critical-and-pseudo-critical-edges-in-minimum-spanning-tree/>

### 10.6 Graph Coloring / Bipartite Check

1. Is Graph Bipartite? — <https://leetcode.com/problems/is-graph-bipartite/>
2. Possible Bipartition — <https://leetcode.com/problems/possible-bipartition/>
3. Flower Planting With No Adjacent — <https://leetcode.com/problems/flower-planting-with-no-adjacent/>



4. Course Schedule (cycle check) — <https://leetcode.com/problems/course-schedule/>

## 10.7 Connected Components / Grid Graphs

1. Number of Connected Components in an Undirected Graph — <https://leetcode.com/problems/number-of-connected-components-in-an-undirected-graph/>
  2. Number of Distinct Islands — <https://leetcode.com/problems/number-of-distinct-islands/>
  3. Making A Large Island — <https://leetcode.com/problems/making-a-large-island/>
  4. Surrounded Regions — <https://leetcode.com/problems/surrounded-regions/>
  5. Pacific Atlantic Water Flow — <https://leetcode.com/problems/pacific-atlantic-water-flow/>
  6. Bricks Falling When Hit — <https://leetcode.com/problems/bricks-falling-when-hit/>
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## Total Summary

- **Categories:** 10
- **Patterns:** 56
- **Total curated LeetCode questions:** 300+ (5-10 per pattern)

## Suggested Order to Solve (Beginner → Advanced)

1. Arrays → Strings → Binary Search
2. Recursion → Linked List
3. Stacks & Queues → Binary Trees & BST
4. Priority Queues (Heaps) → Graphs
5. Dynamic Programming (last, sabse tough)

**Tip:** Har pattern me pehle Easy problem karo (pattern samajhne ke liye), phir Medium (asli interview level), aur last me 1-2 Hard (mastery ke liye). Roz 1 naya pattern + 5 questions ka target rakho — 56 din me pura roadmap complete ho sakta hai.